

EE150 Signal and System

Homework 7

December 18, 2025

Note:

- Please provide enough calculation process to get full marks.
- Please submit your homework to Gradescope in PDF version.
- It's highly recommended to write every exercise on a single sheet of page.
- Late submissions will have points deducted according to the penalty policy.
- Please use English only to complete the assignment, solutions in Chinese are deducted according to the penalty policy.
- Plagiarizer will get zero points.
- The full score of this assignment is 100 points.

Exercise 1. (20pt)

Find the Laplace transform and regions of convergence of the following function:

1. $x(t) = e^{2t}u(-t) + e^{3t}u(-t)$

2. $x(t) = \begin{cases} t, & 0 \leq t \leq 1 \\ 2 - t, & 1 < t \leq 2 \end{cases}$

3. $x(t) = (1 - e^{-4t})u(2t)$

4. $x(t) = t^2u(t - 1)$

Exercise 2. (20pt)

Find the inverse Laplace transform of the following function:
(Hint: discuss different regions of convergence)

1. $X(s) = \frac{3s^2 + s + 1}{(s+2)^2(s+1)} e^{-2s}$

2. $X(s) = \frac{s+1}{(s+1)^2 + 9}$

3. $X(s) = \frac{s^2 + 3s + 3}{s^2 + 2s + 2}$

4. $X(s) = \frac{s}{(s+1)^2}$

Exercise 3. (10pt)

Consider a continuous-time first-order stable LTI system. In the s -domain, the system has a single pole located at $s = -3$, which has no zeros.

- (1) Determine the ROC of $H(s)$, derive the expression of $H(s)$ (assume the gain is 1 for simplicity), and determine whether this system is a causal system.
- (2) Draw the vector from the pole to the complex frequency point $s = j4$. Label the magnitude of this vector (denoted as a) and the angle (denoted as φ) between this vector and the $\text{Re}[s]$ axis.
- (3) Explain the relationship between the magnitude a and angle φ and the value of the system's frequency response $H(j\omega)$ at $\omega = 4$.

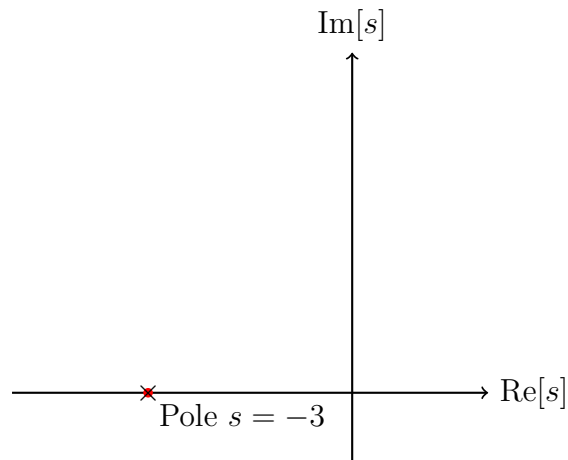


Figure 1: Pole-zero plot in the s -plane

Exercise 4. (15pt)

For a continuous-time system, the differential equation is given by:

$$y''(t) + 7y'(t) + 10y(t) = 2x'(t) + 3x(t), \quad x(t) = e^{-t}u(t)$$

with initial conditions $y(0^-) = 1$, $y'(0^-) = 1$.

- (1) Find the zero-input response.
- (2) Find the zero-state response.
- (3) Find the full response.

Exercise 5. (15pt)

Consider a stable second-order LTI system that satisfies the following condition:

If the input signal is $x(t) = e^t \sin t \cdot u(t)$, the Laplace transform $Y(s)$ of the corresponding output signal $y(t)$ is $Y(s) = \frac{2}{(s+1)^2+1}$;

Answer the following questions:

- (1) Determine $H(s)$ and its region of convergence (ROC).
- (2) Judge whether the system is an all-pass system, and explain the reason.
- (3) Draw the corresponding block diagram of this system.

Exercise 6. (20pt)

Suppose we are given the following information about a causal LTI system S with impulse response $h(t)$ and a rational system function $H(s)$:

1. $h(0^+) = 0.2$.
2. When the input is $u(t)$, the output is absolutely integrable.
3. When the input is $tu(t)$, the output is not absolutely integrable.
4. The signal $\frac{d^2h(t)}{dt^2} + 2\frac{dh(t)}{dt} + 2h(t)$ is a bounded signal that is finite in duration.
5. $H(s)$ has exactly one zero at infinity.

Answer the following questions:

- (1) Determine $H(s)$ and its region of convergence.
- (2) Draw the pole-zero plot and indicate the ROC of $H(s)$.
- (3) Determine whether the system is stable. If so, find out the frequency response $H(j\omega)$.